Velocity graphs and components

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Read a velocity graph and describe motion in two directions.

SUCCESS CRITERIA - I can...

1. Use signed velocity-time area to find displacement.
2. Represent horizontal and vertical components separately.
3. Distinguish graph height, slope and vector magnitude.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Positions are 18, 12, 6 m at 0, 2, 4 s. Predict position at 8 s. Give the rate, an assumption and a future test.

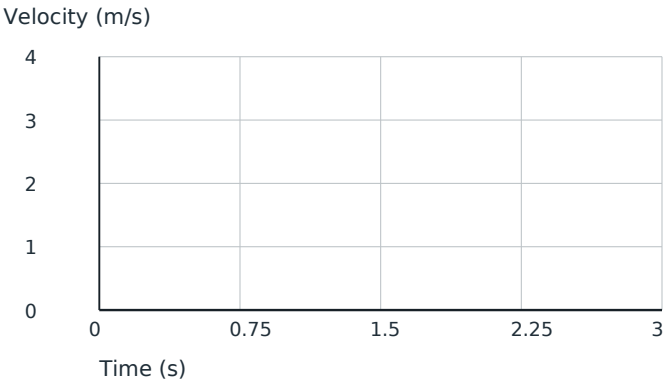
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v_x = v \cos\theta$; $v_y = v \sin\theta$.
θ measured from +x; degree mode. Signed velocity-time area gives displacement; total absolute area gives distance.

Worked reasoning: A cart travels at 3 m/s east for 2 s. The rectangle under its velocity-time graph has area 3×2=6 m, its displacement.

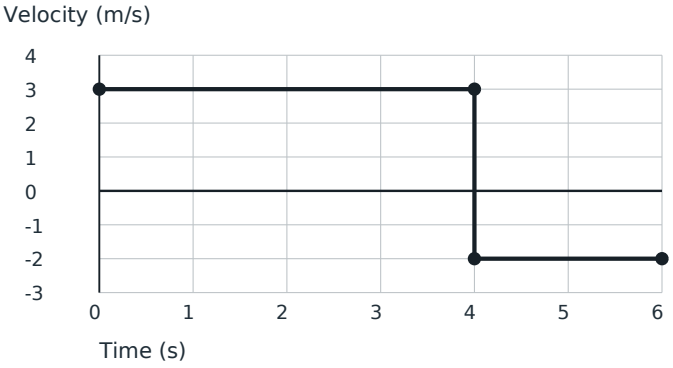
Your turn: Draw the cart's constant velocity on the axes. Shade the area between the line and the time axis. Explain why multiplying the two axis units gives meters.



Marzano check - circle: ① ② ③

03 Add signed areas

Use the graph. Find displacement and distance from 0 to 6 s. Describe the motion on each interval.



Marzano check - circle: ① ② ③

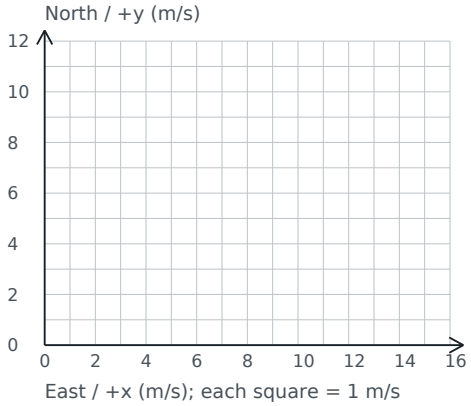
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Continue down each column.

04 Resolve a northeast velocity

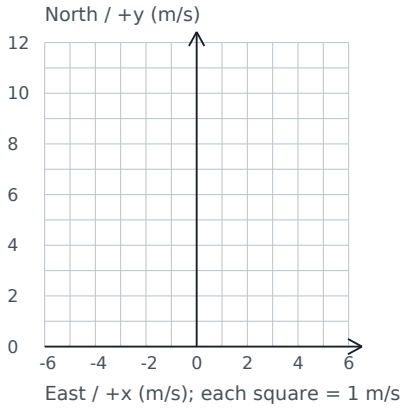
A drone flies at 14 m/s, 35° north of east. Sketch its velocity and component triangle on the grid. Predict which component is larger, then calculate both.



Marzano check - circle: ① ② ③

05 Change the reference axis

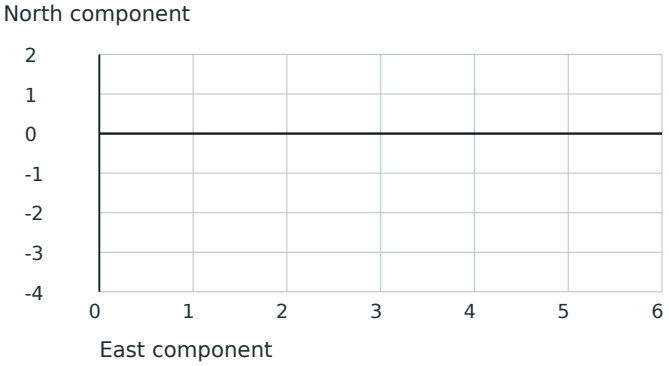
A boat travels at 10 m/s, 20° west of north. East is +x and north is +y. Sketch the velocity, mark the angle from north, and find both signed components. Explain your sine/cosine choices.



Marzano check - circle: ① ② ③

06 Draw velocity and displacement

An object has v=(+4,-3) m/s for 2 s. Draw the velocity vector and calculate its speed, displacement vector, and displacement magnitude.



Marzano check - circle: ① ② ③

07 Two horizontal graphs

Explain the difference between a horizontal line at +5 on a position-time graph and a horizontal line at +5 on a velocity-time graph. Include units.

Marzano check - circle: ① ② ③

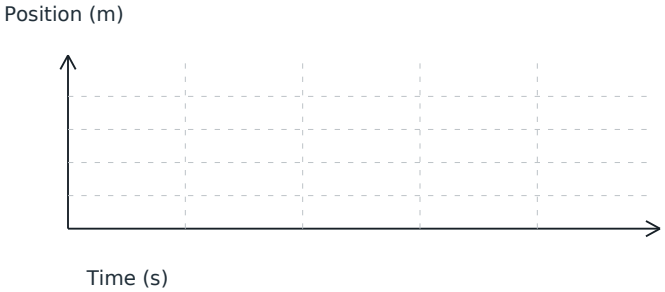
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Continue down each column.

08 Review: connect an earlier idea

A straight position-time segment goes from (1 s, -3 m) to (5 s, 17 m). Find velocity, interpret its units and predict x at 7 s if motion continues.
Label the model with the given positions and any prediction.



Marzano check - circle: 1 2 3

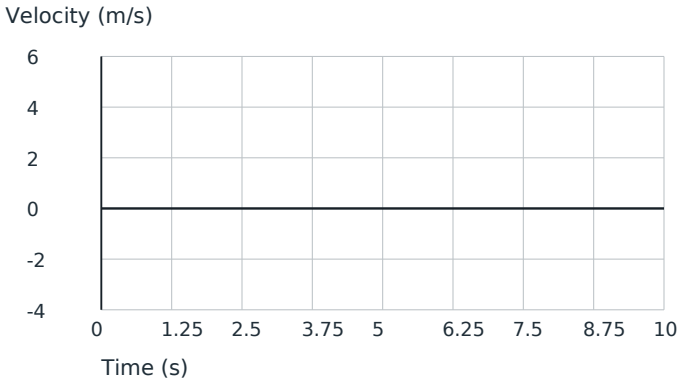
09 Exit check

A velocity graph stays at +2 m/s for 3 s, then -1 m/s for 2 s. Find displacement and distance. State the information needed to interpret a vector angle of 30°.

Marzano check - circle: 1 2 3

10 STRETCH: test the model

A cart moves at +4 m/s for 3 s, then -2 m/s. How long must the second interval last to return to its start? Sketch the graph and find average speed for the full trip.



Marzano check - circle: 1 2 3